

Taiki Ishigaki

Ph.D. in Information Science and Technology | Assistant Professor, Tokyo University of Science

Research interests: hybrid-link system dynamics, differentiable simulation, distance fields in configuration spaces, humanoid control, soft robotics, and human motion analysis for flexible embodied systems.

RESEARCH KEYWORDS

Hybrid-link system

Config-space distance fields

Humanoid control

Soft robotics

Human motion analysis

CURRENT AFFILIATION

Tokyo University of Science, Research Institute for Science and Technology
Department of Functional Design Engineering, Faculty of Advanced Engineering
[Interactive Robotics Laboratory](#)

POSITION

Assistant Professor

E-MAIL

taiki000ishigaki [at] gmail.com

CAREER

Academic Appointments

Apr. 2025 - Present	Assistant Professor Tokyo University of Science
Apr. 2024 - Mar. 2025	Project Researcher The University of Tokyo
Aug. 2024 - Mar. 2025	Collaborative Researcher AIST
Feb. 2021 - Mar. 2021	Technical Research Assistant The University of Tokyo

EDUCATION

Education

Apr. 2021 - Mar. 2024	Doctoral Program, Ph.D. in Information Science and Technology The University of Tokyo
Apr. 2019 - Mar. 2021	Master's Program The University of Tokyo
Apr. 2015 - Mar. 2019	Bachelor's Program Yokohama National University / Materials Engineering major, Robotics and Mechatronics minor

SELECTED WORKS

Selected Publications and Book

- Taiki Ishigaki**, Ko Ayusawa, Ko Yamamoto, and Eiichi Yoshida, "[Expanded Motion Computation Theory: Higher-Order Differential Kinematics from Rigid-Link Systems to Continuous Flexible Rods](#)", Journal of the Robotics Society of Japan, 2026. [in Japanese]
- Taiki Ishigaki**, Teresa Vidal-Calleja, Ko Ayusawa, and Eiichi Yoshida, "Implicit Null-space Manifold Generation for Redundant Robotic Systems," RSS 2026, accepted.
- Sunghee Kim, Yuta Shimane, **Taiki Ishigaki**, and Ko Yamamoto, "[Dynamics computation of soft-rigid hybrid-link system and its application to motion analysis of an athlete wearing sport prosthesis](#)", Advanced Robotics, 2026.
- Taiki Ishigaki**, Ko Ayusawa, and Ko Yamamoto, "[Unified Framework of Gradient Computation for Hybrid-Link System and Its Dynamical Simulation by Implicit Method](#)", IEEE Robotics and Automation Letters, 2025.
- Taiki Ishigaki**, Ko Ayusawa, and Ko Yamamoto, "[Comprehensive Gradient Computation Framework of PCS Model for Soft Robot Simulation](#)", IEEE Robotics and Automation Letters, 2024.
- Ko Yamamoto, **Taiki Ishigaki**, and Yoshihiko Nakamura, "[Humanoid Motion Control by Compliance Optimization Explicitly Considering its Positive Definiteness](#)", IEEE Transactions on Robotics, 2022.
- Taiki Ishigaki** and Ko Yamamoto, "[Dynamics Computation of a Hybrid Multi-link Humanoid Robot Integrating Rigid and Soft Bodies](#)", IROS 2021.
- Ko Yamamoto, Ko Ayusawa, and **Taiki Ishigaki**, *Robot Motion Computation Theory*, Corona Publishing, Apr. 2026. Textbook contribution on motion computation for robotic systems. [in Japanese]

RECOGNITION

Awards and Honors

- 2026**: Excellent Paper Award, 31st Robotics Symposia, for higher-order differential kinematics and optimization of multi-link systems.
- 2024**: ROBOMECH Award, Academic Research Category, for high-speed forward dynamics simulation of hybrid rigid-soft link systems.
- 2023**: Outstanding Research and Technology Award, Robotics Society of Japan, for PCS-model-based deformation analysis and viscosity estimation of sports prostheses.

FUNDING

Grants and Fellowships

- 2025-2028**: JSPS KAKENHI Early-Career Scientists, supporting differential motion computation theory for dynamic identification of hybrid-link systems.
- 2025-2026**: Foundation for the Fusion of Science and Technology, supporting data assimilation methods for simulation of flexible sports tools.
- 2024-2025**: [Daiichi-Sankyo "Habataku" Support Program](#), supporting differentiable hybrid-link simulation for humans and soft-rigid robots.
- 2021-2024**: International Graduate Program of Excellence for an Intelligent Society Fellowship, supporting doctoral research at The University of Tokyo.

COMMUNITY

Academic Service

- Associate Editor: Humanoids 2025; Organizing Committee Member: HVAC2026.
- Reviewer: ICRA, IROS, Humanoids, ISRR, SII, RA-L, T-RO, and Advanced Robotics.
- Professional memberships: Robotics Society of Japan, Society of Instrument and Control Engineers, Japan Society of Mechanical Engineers, and IEEE.